Design Retrospective — Patient Observation Tracker

CSCI-P532 Object Oriented Software Development · Spring 2026 · Week 2 · Rohith Gowda Devaraju · rdevaraj@iu.edu

Change 1 — Multiple Diagnosis Strategies

Status	File
NEW	`WeightedScoringStrategy`
NEW	`DiagnosisStrategyConfig` (strategy Map bean)
NEW	`StrategyType` enum, `EvaluationResult` DTO
MOD	`DiagnosisEngine` — dispatches by `rule.getStrategyType()`
MOD	`SimpleConjunctiveStrategy` — filters INFERRED obs
MOD	`AssociativeFunction` — added strategy, weights, threshold

Untouched: ObservationManager, ObservationFactory, all Commands, all Decorators, all Listeners.

> **Unexpected touch:** None. Strategy pattern absorbed the change entirely within the engine layer.

Change 2 — Decorator Processing Pipeline

Status	File
NEW	`UnitValidationDecorator`
NEW	`AnomalyFlaggingDecorator`
NEW	`AuditStampingDecorator`
NEW	`BaseObservationProcessor`
MOD	`ObservationProcessorConfig` — assembles chain
MOD	`ObservationManager` — calls `processor.process()`
MOD	`ObservationFactory` — unit validation removed

Untouched: DiagnosisEngine, all Strategies, all Commands, all Listeners.

> **Unexpected touch — `ObservationFactory`:** Unit validation had to be removed from the factory and moved into `UnitValidationDecorator`. *Redesign:* The factory should only construct objects; validation should have always lived in a separate collaborator, so no factory change would be needed when decorators change.

Change 3 — Multi-User Support & Command Undo

Status	File
NEW	`AuditableCommandLog` — reads real user from context
NEW	`UndoManager` — enforces same-user, single-undo
NEW	`ObservationAwareCommand` interface
MOD	`CommandLog` — captures observationId, reads user context
MOD	`RecordObservationCommand` — added `undo()` → REJECTED
MOD	`RejectObservationCommand` — added `undo()` → ACTIVE
MOD	`CreatePatientCommand` — added `undo()` → throws UnsupportedOp

Untouched: ObservationFactory, DiagnosisEngine, all Strategies, all Decorators, all Listeners.

> **Unexpected touch:** None. Command interface extension (adding `undo()`) was confined to the command layer. `AuditableCommandLog` extended rather than modified the original, preserving OCP.

Change 4 — Concept Hierarchy & Propagation

Status	File
NEW	`PropagationListener` — infers PRESENT ancestors / ABSENT descendants
NEW	`ObservationSource` enum (MANUAL / INFERRED)
MOD	`RuleEvaluationListener` — filters MANUAL obs only
MOD	`DiagnosisEngine` — pre-filters to MANUAL before dispatch
MOD	`ObservationFactory` — added `createInferred()` variant
MOD	`PhenomenonTypeManager` — resolves parentConcept FK

Untouched: AuditLogListener, all Commands, CommandLog, ObservationManager, all Decorators.

> **Unexpected touch — `DiagnosisEngine`:** Touched to pre-filter INFERRED obs before any strategy runs. *Redesign:* An injectable `ObservationFilter` predicate passed into the engine at construction time would eliminate future engine changes when filtering rules evolve.

Summary — Pre-existing Pattern-Layer Files Modified per Change

Change	New Files	Pre-existing Modified
C1 — WeightedScoringStrategy + strategy selection	4	2
C2 — Decorator pipeline (3 decorators)	4	3
C3 — Multi-user + Command undo	3	4
C4 — Concept hierarchy + propagation listener	2	4